

Door Unlocking System Using RFID

Arduino/NodeMCU ESP8266 source code

```
#include <SPI.h>
#include <MFRC522.h>
#include <Servo.h>

#define SS_PIN D2
#define RST_PIN D1
#define SERVO_PIN D4
#define BUZZER_PIN D0

MFRC522 rfid(SS_PIN, RST_PIN);
Servo doorServo;

// Replace this UID with your RFID card's UID
byte authorizedUID[] = {0xA1, 0xB2, 0xC3, 0xD4};

void setup() {
  Serial.begin(115200);

  SPI.begin();
  rfid.PCD_Init();

  doorServo.attach(SERVO_PIN);
  doorServo.write(0); // Door locked

  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);

  Serial.println("RFID Door Lock System");
  Serial.println("Scan your RFID card...");
}

void loop() {

  // Check whether a new RFID card is present
  if (!rfid.PICC_IsNewCardPresent()) {
    return;
  }

  // Read the RFID card
  if (!rfid.PICC_ReadCardSerial()) {
    return;
  }

  Serial.print("Card UID: ");

  for (byte i = 0; i < rfid.uid.size; i++) {
    Serial.print(rfid.uid.uidByte[i], HEX);
    Serial.print(" ");
  }

  Serial.println();

  // Check authorized card
  if (checkUID()) {
    Serial.println("Access Granted");

    digitalWrite(BUZZER_PIN, HIGH);
    delay(200);
    digitalWrite(BUZZER_PIN, LOW);

    // Unlock door
    doorServo.write(90);
    delay(5000);

    // Lock door
    doorServo.write(0);

    Serial.println("Door Locked");
  }
  else {
    Serial.println("Access Denied");

    // Buzzer for invalid card
    digitalWrite(BUZZER_PIN, HIGH);
    delay(1000);
    digitalWrite(BUZZER_PIN, LOW);
  }
}
```

```
rfid.PICC_HaltA();
rfid.PCD_StopCrypto1();

delay(1000);
}

// Function to compare RFID UID
bool checkUID() {

    if (rfid.uid.size != sizeof(authorizedUID)) {
        return false;
    }

    for (byte i = 0; i < rfid.uid.size; i++) {
        if (rfid.uid.uidByte[i] != authorizedUID[i]) {
            return false;
        }
    }

    return true;
}
```